Emergency Maintenance Report — EMR-2025-006

APEX EMERGENCY GRID SOLUTIONS
24/7 High-Voltage Rapid Response · NAESB-Certified Emergency Contractor
1450 Industrial Parkway, Tacoma, WA 98421 · Dispatch: (253) 555-0177

Field	Value
Document ID	EMR-2025-006
Issued By	Apex Emergency Grid Solutions
Report Date	2025-12-22
Site	NW-PowerPool-WIND-003
Asset	TRA-10561 (Siemens Energy power transformer, 138 kV / 34.5 kV, 75 MVA)
Outage ID	OUT-2025-1407
Cascadia Work Order	WO-2025-12041
Cascadia Authorization	EPA-2024-11 (invoked 2025-12-10 01:24 PST)
Apex Crew Lead	M. Castellanos, Lead Transformer Specialist
OEM Liaison	Siemens Energy — H. Brandt, Transformer Field Engineering
Distribution	Cascadia Operations · Cascadia Engineering · Cascadia Vendor Management · Cascadia AP

1. Scope of Emergency Work

REPEAT-ASSET ALERT. Apex was dispatched at 01:24 PST 2025-12-10 in response to an overheating trip on TRA-10561. **This is the second Apex emergency deployment to TRA-10561 in FY2025** (prior event 2025-05-26, OUT-2025-1405, Apex report EMR-2025-002). The temporary winding restraint and derated configuration installed under EMR-2025-002 have not held.

CPS reported >8-hour mobilization (winter weather across the Cascades).
EPA-2024-11 invoked.

Authorized scope:
- Onsite isolation, oil sampling, full electrical and acoustic assessment.
- Stabilization to whatever degree possible.
- OEM-supervised assessment for capital scope.

Outage duration to restoration: **215 hours**. Lost generation per Cascadia SCADA: 3,558.6 MWh.

2. Findings

Onsite arrival 2025-12-10 09:46 PST (delay due to snow on access road).

1. **DGA on main tank oil**: acetylene 41 ppm (up from 14 ppm in May), ethylene 178 ppm (up from 92), hydrogen 410 ppm (up from 280). IEEE C57.104 places the unit firmly in **Condition 4 — take unit out of service**. The May intervention bought time; it did not arrest the degradation.
2. **Acoustic emission survey** confirmed partial discharge activity has progressed from the HV winding outboard end (May finding) to involve the inboard end as well, on phases A and B.
3. **Vibration spectrum at the LV bushing turret** now showing 120 Hz at 24.1 mm/s — well past the ISO 10816-3 Zone D threshold and substantially worse than May (18.4 mm/s). The mechanical wedging installed under EMR-2025-002 has loosened.
4. **Tank top temperature** reached 118 °C against an alarm threshold of 105 °C immediately prior to trip.
5. The winding inter-turn insulation degradation flagged in EMR-2025-002 has advanced. In Apex's judgment, supported by the on-site Siemens Energy engineer, **continued operation of TRA-10561 in any configuration carries unacceptable risk of catastrophic failure with collateral damage to adjacent assets**.
6. **Apex cites prior emergency work (EMR-2025-002) as direct evidence that field-stabilization scope is no longer viable and that imminent failure is the operating assumption.**

3. Repairs Performed

Emergency stabilization to permit safe de-energization and out-of-service condition. **The unit was not returned to service under this scope.**

Item	Detail	Hours / Qty
Snow road clearance, site access	Subcontractor, half day	8 hrs
De-energization, lock-out/tag-out	Lead + 2 technicians	14 hrs
DGA dispatch, expedited lab	Acetylene, ethylene, hydrogen suite	6 hrs
Acoustic emission and vibration survey	Full enclosure, 3 passes	38 hrs
Tank oil partial drawdown, conditioning, refill	1,800 gal	44 hrs
Mechanical wedging re-tension	Confirmed inadequate; documented	22 hrs
Stepped energization attempts	3 attempts, all aborted on PD signature	28 hrs
OEM (Siemens Energy) on-site engineering coordination	H. Brandt, 2 days	18 hrs
Weekend / off-hours, winter conditions	1.5× multiplier	37 hrs
Total labor (Apex + OEM pass-through)		**215 hours**

Materials: 1,800 gallons mineral oil; mechanical wedging refresh kit; DGA lab consumables.

4. Recommendations

In Apex's professional judgment, supported by Siemens Energy on-site engineering and consistent with our position in EMR-2025-002:

1. **Full transformer replacement is required immediately.** Field repair is no longer credible; the May 2025 stabilization (EMR-2025-002) demonstrably did not arrest degradation. **The prior emergency work itself is evidence of imminent failure.** Apex recommends Cascadia procure a replacement unit on an expedited basis. Estimated capital scope, full replacement: **$2.85M** (Siemens Energy quote SE-Q-2025-414, including new transformer, freight, foundation work, and commissioning).
2. **TRA-10561 should not be re-energized in any configuration** pending the replacement. Apex cannot warrant any re-energization scope on this asset.
3. The cumulative emergency-stabilization spend on TRA-10561 in FY2025 is approximately $791K (EMR-2025-002 + EMR-2025-006), with no return-to-service capability remaining. The replacement decision is, in Apex's view, overdue.

4. Apex notes Cascadia internal MOC-2025-022 is pending engineering review. Apex's report is bound to OEM and observed-condition evidence; resolution of the internal MOC is outside our scope.

5. Invoice Summary

Line	Rate	Qty	Extended
Labor — emergency rate, standard hours	$420.00/hr	138 hrs	$57,960.00
Labor — emergency rate, weekend / winter 1.5×	$630.00/hr	59 hrs	$37,170.00
Labor — OEM Siemens Energy pass-through (subcontract 1.8×)	$756.00/hr	18 hrs	$13,608.00
Subtotal labor			$108,738.00
Parts — mineral oil, 1,800 gal	$11.40/gal	—	$20,520.00
Parts — wedging kit, gaskets, consumables	itemized	—	$11,800.00
Emergency parts mark-up	35% on parts subtotal	—	$11,312.00
OEM engineering review	flat	—	$36,000.00
DGA laboratory, expedited	flat	—	$5,400.00
Snow clearance / access subcontract	pass-through + 15%	—	$14,800.00
Crane / lift services (subcontract)	pass-through + 12%	—	$32,400.00
Travel, per-diem (3 Apex + 1 OEM × 9 days, winter)	per Apex tariff	—	$19,800.00
Mobilization premium (EPA <4hr dispatch, winter, off-hours)	flat	1	$62,000.00
EPA-2024-11 emergency response surcharge	30% on Apex labor	—	$28,539.00
Documentation, OEM-format condition report	flat	—	$8,400.00
Subtotal			**$369,709.00**
	—	—	$32,291.00

Line	Rate	Qty	Extended
Risk / availability premium (per Apex MSA §4.3)			
Total Invoice			**$402,000.00**

Note: This invoice does **not** include the proposed full replacement scope ($2.85M, separate Siemens Energy quote SE-Q-2025-414).

Closing

TRA-10561 is in out-of-service, de-energized condition as of 2025-12-19 at 04:12 PST. Apex will not authorize re-energization absent a full replacement. We urge Cascadia engineering to expedite the replacement decision.

Signed	Name	Date
Crew Lead	M. Castellanos	2025-12-22
Apex Engineering Review	D. Pieterson, PE	2025-12-22
OEM Field Engineer (concurrence)	H. Brandt (Siemens Energy)	2025-12-20
Apex Account Manager	J. Beaumont	2025-12-22

Attachments: DGA lab report (expedited) · acoustic emission traces · vibration spectra (12) · thermal imaging · Siemens Energy quote SE-Q-2025-414 · prior report EMR-2025-002 referenced.
Apex references: Siemens Energy Manual SE-TR-2300 · IEEE C57.104 · ISO 10816-3 · Apex SOP AEX-SOP-XFR-04.
Cascadia references cited in dispatch packet: EPA-2024-11 · MOC-2025-022 · CAR-2025-002 · RCA-2025-002 · prior report EMR-2025-002.